

G. J. Boer · G. Flato · M. C. Reader · D. Ramsden

A transient climate change simulation with greenhouse gas and aerosol forcing: experimental design and comparison with the instrumental record for the twentieth century

Received: 24 September 1998 / Accepted: 8 October 1999

Abstract The Canadian Centre for Climate Modelling and Analysis (CCCma) global coupled model is used to investigate the potential climate effects of increasing greenhouse gas (GHG) concentrations and changes in sulfate aerosol loadings. The forcing scenario adopted closely resembles that of Mitchell et al. for both the greenhouse gas and aerosol components. Its implementation in the model and the resulting changes in forcing are described. Five simulations of 200 years in length, nominally for the years 1900 to 2100, are available for analysis. They consist of a control simulation without change in forcing, three independent simulations with the same greenhouse gas and aerosol changes, and a single simulation with greenhouse gas only forcing. Simulations of the evolution of temperature and precipitation from 1900 to the present are compared with available observations. Temperature and precipitation are primary climate variables with reasonable temporal and spatial coverage in the observational record for the period. The simulation of potential climate change from the present to the end of the twenty-first century, based on projected GHG and aerosol forcing changes, is discussed in a companion paper. For the historical period dealt with here, the GHG and aerosol forcing has changed relatively little compared to the forcing changes projected to the end of the twenty-first century. Nevertheless, the forced climate signal for temperature in the model is reasonably consistent with the observed global mean temperature from the instrumental record. This is true also for the trend in zonally averaged temperature as a function of latitude and for some aspects of the geographical and regional distributions of temperature. Despite the modest change in overall forcing, the difference between GHG + aerosol and GHG-only forcing is discernible in the temperature response for this period. Changes in

precipitation, on the other hand, are much less evident in both the instrumental and simulated record. There is an apparent increasing trend in average precipitation in both the observations and the model results over that part of the land for which observations are available. Regional and geographical changes and trends (which are less affected by sampling considerations), if they exist, are masked by the large natural variability of precipitation in both model and observations.

1 Introduction

The simulation of time-dependent historic and projected climate change with reasonably complete coupled climate models has a comparatively short history. The evolution of coupled models and their use for simulating potential climate change is reflected in the various reports of the Intergovernmental Panel on Climate Change (referred to here as IPCC 1990, 1992, and 1995). In IPCC 1990, climate change simulations are equilibrium simulations where relatively complete and complex atmospheric general circulation models are coupled to a simple ocean component usually consisting of a 50 m ocean mixed layer. An earlier version of the CCCma climate model (McFarlane et al. 1992; Boer et al. 1992) is of this kind. The temperature of the ocean mixed layer is regulated by the exchange of heat with the atmosphere via radiative, sensible and latent heat fluxes, and by an imposed and non-interactive geographically distributed source/sink of heat, R . Ideally R represents the horizontal and vertical oceanic heat transport into and out of the ocean surface layer, but it also provides a “flux adjustment” which helps ensure that the current climate is well simulated over the oceans. Since R does not interact with the atmosphere, or the rest of the ocean, the implied transport of heat does not change in the climate change calculations made with these models.

Despite the simplicity of these earlier coupled models, their behaviour gives an estimate of the first order climate change expected with a change in external forcing.

G. J. Boer (✉) · G. Flato · M. C. Reader · D. Ramsden
Canadian Centre for Climate Modelling and Analysis,
Atmospheric Environment Service,
University of Victoria, B.C. Canada
E-mail: George.Boer@ec.gc.ca

A “first standard” climate change experiment consists of the simulation of the current or $1 \times \text{CO}_2$ climate together with the simulation of an equilibrium $2 \times \text{CO}_2$ climate, where the concentration of CO_2 in the model atmosphere has been doubled. Simulated temperature changes show characteristic features with, for instance, the largest warming found at high latitudes, over the continents, and in winter.

Analyses of equilibrium climate change point out the strengths and some weaknesses of the climates simulated with this generation of models. In particular, projections of the time evolution of climate change require a more sophisticated approach which includes the interaction between the surface and deep ocean as well as changes in ocean transport. More complete coupled models, where the atmospheric GCM is coupled to a full three-dimensional ocean GCM, are used for this purpose. A “second standard” climate change experiment consists of a control simulation with a fully coupled model for “present” or $1 \times \text{CO}_2$ forcing (typically that for the 1980s) together with a transient simulation in which the CO_2 concentration is increased by 1% per year from the present value to, and beyond, the time of CO_2 doubling at about 2050. A range of results from simulations of this kind are discussed in Chapters 5 and 6 of IPCC 1995. The Coupled Model Intercomparison Project (Meehl et al. 1997) is an international effort to analyze and intercompare control and forced simulations of this kind. Difficulties with this experiment include model initialization and the use and necessity of flux adjustment to minimize climate drift.

Finally, both the first and second standard experiments give results which imply a stronger warming than is easily reconciled with the observed climate record. This, together with the recognition that direct and possibly indirect effects of sulfate aerosols would act to counteract the greenhouse gas warming, leads to a potential “third standard experiment” the results of which are discussed here for the CCCma coupled model. This experiment consists of several long simulations nominally from “preindustrial” 1850 conditions to the end of the twenty-first century. A representation of the historical change in greenhouse gas (GHG) and aerosol forcing from 1850 to the present is specified in terms of equivalent CO_2 concentration and aerosol loading changes. The projected forcing change from the present to 2100 is a modified version of the IPCC 1992a scenario and consists of a 1% per year increase in CO_2 concentration and an associated change in aerosol loading.

There are a number of advantages to this general approach: (1) initialization transients (such as the cold-start) are avoided; (2) GHG and direct sulfate aerosol forcings are both considered; (3) the model’s ability to simulate aspects of the historic climate record may be evaluated; (4) the effect of the aerosol forcing may be estimated and investigated; and (5) the several realizations of the transient evolution allow better separation of the forced climate signal from the internal variability of the system.

In what follows, the coupled model, its initialization, and its control climate are reviewed. The historical and projected forcing scenario and its implementation in the model is discussed, and the distribution in time and space of the resulting GHG and aerosol forcing in the model is analyzed. Finally, the simulations of temperature and precipitation from 1900 to the present are analyzed and compared with available observations.

2 The coupled model

The CCCma global coupled model (CGCM1) and its control simulation are discussed in a companion paper (Flato et al. 2000). Basic features of the model are summarized here.

2.1 The atmospheric component

The atmospheric component of the coupled model is the CCC second generation AGCM described in McFarlane et al. (1992). It is a spectral model with T32 triangular resolution corresponding to a transform grid of 96×48 “Gaussian” latitudes or approximately 3.75° in latitude and longitude. There are 10 levels in the vertical. The model has a comparatively sophisticated representation of radiative transfer (including annual and diurnal cycles), surface heat and moisture budgets, and clouds and cloud optical properties. Results from this version of the AGCM appear in the various publications associated with the first AMIP intercomparison (extensive references and other information can be found on the AMIP web site).

This is also the version of the atmospheric model that is coupled to a simple mixed layer thermodynamic ocean and thermodynamic sea-ice model for the “first standard” or $2 \times \text{CO}_2$ equilibrium climate change experiment, results of which appear in IPCC 1990. Boer et al. (1992) analyze the simulated $2 \times \text{CO}_2$ equilibrium climate change in some detail. The results of this simulation are used in a variety of other studies; Boer (1993) discusses changes in the model’s hydrological cycle and surface energy budgets; Lambert (1995) analyzes the cyclone climatology in the $1 \times \text{CO}_2$ and $2 \times \text{CO}_2$ simulations; Boer (1995) shows that, despite the considerable change in surface climate and a modest decrease in the basic “rate of working” of the $2 \times \text{CO}_2$ atmosphere (the rate of generation of available potential energy, conversion to kinetic energy, and subsequent dissipation), the overall poleward transport of energy changes little since a decrease in internal energy transport is compensated by an increase in latent energy transport. Zwiers and Kahren (1998) investigate changes in extremes and return periods for temperature and precipitation for the $2 \times \text{CO}_2$ climate.

Finally, Reader and Boer (1998) consider the direct effect of sulfate aerosols in an equilibrium $2 \times \text{CO}_2 + \text{aerosol}$ simulation which has relevance to the transient simulations discussed here. They argue that, despite the comparatively localized nature of the radiative forcing associated with the sulfate aerosols, the first order climate response is nevertheless to counteract the (non-localized) GHG forcing. The patterns of response are largely determined by the feedbacks in the system (e.g. land versus ocean and the temperature/albedo effects) rather than the geographical structure of the forcing itself.

2.2 The oceanic component

The oceanic component of the coupled model is a version of the GFDL MOM model (Pacanowski et al. 1991) at a comparatively high resolution of $1.8^\circ \times 1.8^\circ \text{L29}$. For each grid box of the AGCM’s transform grid of 96×48 gaussian grid points there are four oceanic grid boxes. The coupling between components occurs once per simulated day with the daily average heat and fresh water fluxes (plus their flux adjustments) and surface stresses (unadjusted)

from the AGCM supplied to the ocean component which is integrated for one day. The change in sea-surface temperature thus simulated is supplied to the AGCM which is integrated for one day with SSTs which are interpolated from the current value to this new value.

2.3 Initialization and the control climate

The spin-up, flux adjustment, adaption, and control climate for this first version of the CCCma coupled climate model are discussed in Flato et al. (2000). The coupled model is initialized as follows: the ocean model is separately “spun-up” for over 4000 simulated years to attain equilibrium; the atmospheric model is separately “spun-up” for 20 years; the two components are coupled using flux adjustment; and the coupled model is integrated for a further 14 year “adaption” period where the flux adjustments are further refined. The subsequent coupled model simulations nominally begin at 1850 but data is retained and analyzed for the period 1900–2100 to minimize the effect of initial adjustments and transients. The 200 year control experiment, analyzed in Flato et al. (2000), is also part of CMIP and forms the basis against which to discuss the climate change simulations described later.

3 Transient forced climate change simulations

In addition to the control simulation, three independent transient climate change simulations are performed with GHG + aerosol forcing. A final simulation is performed with GHG-only forcing. The three realizations of the evolution of climate change with the same GHG concentrations and aerosol forcing are undertaken in order to better distinguish the forced climate change from the natural variability of the system. The first of these simulations is begun at the nominal date of 1850 with initial conditions from the control simulation and with the specified GHG concentrations and aerosol loadings described later. The simulation proceeds to 2100 with analysis data retained from 1900–2100. The GHG-only simulation proceeds in the same fashion, with the same GHG concentrations but without aerosol effects.

The remaining GHG + aerosol simulations are initialized by combining the ocean conditions of 1 January, 1889, from the first simulation with atmospheric conditions from a year previously and subsequently, namely from 1 January, 1888, and 1 January, 1890, of the simulation. The perturbed initial states rapidly decouple for the atmosphere and the upper ocean (Boer 2000) so that for the data retention period of 1900–2100 the three GHG + aerosol simulations are effectively independent realizations of the climate evolution under the same forcing. In addition to the five simulations described two “stabilization” integrations are underway where the GHG + aerosol forcing becomes constant at the year 2050 and 2100 values although they are not discussed here.

3.1 Forcing scenarios

The effective CO₂ and aerosol evolution used in these simulations follows that of Mitchell et al. (1995) and is a modified version of the IPCC 1992a scenario (IPCC 1995). The use of a common forcing prescription means that differences in simulated climate change obtained with different models may be ascribed to differences in model response rather than to a combination of both forcing and response. The GHG concentration used follows the observed “effective CO₂ concentration” to 1990 at which point a 1%/year increase scenario is adopted. The annual average aerosol loading patterns are based on the slow-oxidation version of the chemistry model of Langner and Rhode (1991) with the addition of some new Southern Hemisphere sources. The annual average aerosol loading patterns used are zero at 1850, follow the estimated increase to 1986 and are subsequently approximated by a linear combination of the 1986 and 2050 patterns shown in Fig. 1.

The direct effect of single scattering by an optically thin layer of non-absorbing aerosols near the Earth’s surface is simulated, as described in Reader and Boer (1998), by adjusting the surface albedo, α_s , as $\hat{\alpha}_s = \max(\alpha_s + \Delta\alpha_s, 1)$ with $\Delta\alpha_s = \beta\delta(1 - \alpha_s)^2 \sec\theta_{eff}$. The upward scattering parameter, $\beta = 0.29$, is averaged over the solar spectrum and zenith angle. $\hat{\alpha}_s$ is the modified surface albedo and $\delta = (8 \text{ m}^2/\text{g}) B(\text{SO}_4^{2-})$ is the aerosol optical depth which is taken to be proportional to the mass aerosol loading, $B(\text{SO}_4^{2-})$ (with the coefficient appropriate for an average low-level humidity of 80%). θ_{eff} is an effective solar zenith angle which includes the effect of diffuse radiation due to clouds.

The forcing prescription indicates how GHG and aerosol effects are introduced into the model. The resulting radiative perturbations in the model depend on the distribution of temperature, moisture, cloud, and other parameters. The direct radiative effect of GHG concentration and aerosol loading, that is the radiative effect that would occur without any feedback and alteration of the model climate, is obtained diagnostically by integrating the model for one year under preindustrial conditions. Radiative calculations are made with altered GHG concentrations and aerosol forcing but without allowing the results to influence the model evolution. These calculations are performed at 50 year intervals for the years 1900, 1950, 2000, 2050, and 2100.

The aerosol forcing is indicated by shortwave flux differences at the top of the atmosphere, and the GHG forcing by longwave flux differences at the model level $\eta = 0.16$ (approximately 160 hPa) which is taken to approximate the location of the tropopause. Single month tests show that the globally averaged values of these forcing quantities are relatively insensitive to the details of the calculation (aerosol forcing differs by only 2% between the top-of-the-atmosphere and the $\eta = 0.16$ level, while the GHG forcing differs by 4% between the $\eta = 0.16$ and $\eta = 0.265$ levels). Time series of the globally averaged forcing due to CO₂, aerosols, and the total are given in Fig. 2. The GHG forcing increases throughout the simulation, while the aerosol forcing is projected to level off after 2050.

An example of the annual cycles of the global mean GHG and aerosol forcing are shown in Fig. 3 for the year 2000. The variation from month to month is the result of the annual cycle of the model’s

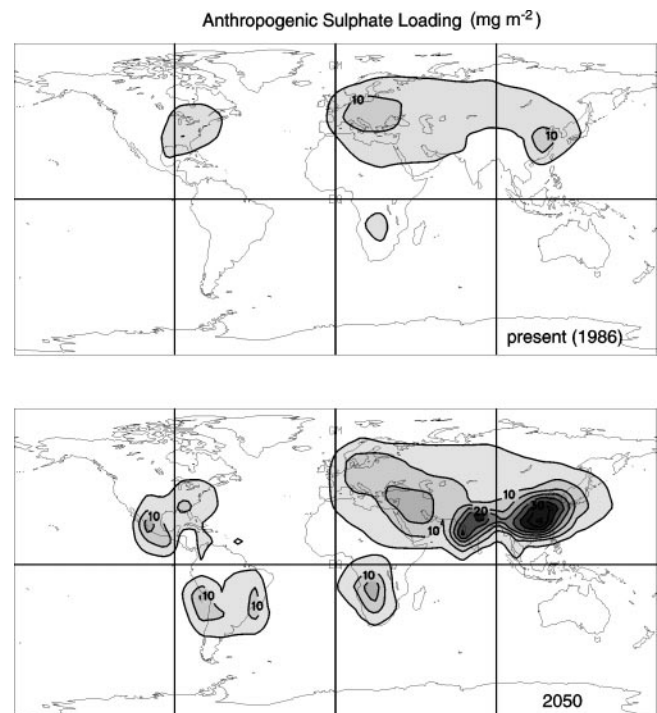


Fig. 1 Anthropogenic sulfate aerosol loading for the *present* (1986) and 2050. Units mg m^{-2}

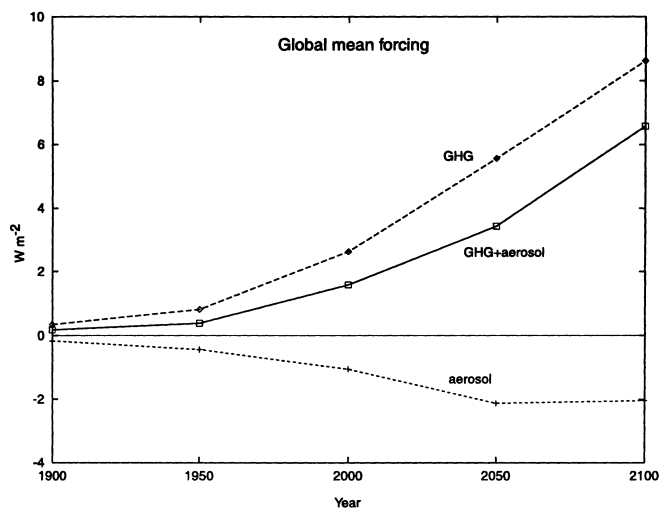


Fig. 2 Global mean *GHG*, *aerosol* and net “direct” forcing changes (without feedback on the climate) as calculated for the control run distributions of temperature, moisture, cloudiness, etc. Units Wm^{-2}

climate variables and not of the *GHG* concentrations or aerosol loadings which do not have an annual cycle in their specification. The seasonal variation of temperature, moisture, and cloudiness for a particular year has little overall effect on the *GHG* forcing which displays a very weak annual cycle. The direct aerosol forcing shows a larger annual cycle with its maximum magnitude occurring in the JJA season in association with the increased length of day in the Northern Hemisphere where the bulk of the aerosol is located.

The geographical distribution of the annual average direct *GHG* and aerosol forcing patterns are shown in Figs. 4 and 5 for 1950 and 2050 (note the different contour intervals for the different times). The *GHG* forcing is geographically very homogeneous and the pattern also scales smoothly in time. The aerosol forcing, on the other hand, is much more localized with a maximum in Asia which increases strongly and Southern Hemisphere regions which become relatively more important at later times. The local nature of the aerosol forcing compared to the *GHG* forcing suggests that a clear signal in the climate response might be expected. However, Reader and Boer (1998) argue, in the context of an equilibrium simulation, that the first order response to the aerosol forcing is not localized but distributed and acts to oppose the response to the broad *GHG* forcing.

3.2 Simulation data and data availability

A rather complete history of both the atmosphere and ocean is retained during these simulations. A subset of the data is available for applications and other investigations via the internet at <http://www.cccma.bc.ec.gc.ca>. These data consist of global grid point arrays of monthly mean surface fields including temperature, precipitation, mean sea-level pressure, and other surface quantities as well as various free atmosphere quantities for each month of the 200 years of the control, the three *GHG* + aerosol simulations, and the *GHG*-only simulation. More details are provided in a companion paper Boer et al. (2000), which discusses the simulation of climate change from the present to the year 2100.

The “historical” simulations reported here attempt to compare simulated climate evolution with available observations for the twentieth century. The analysis concentrates on surface air temperature and precipitation as primary climatic variables with reasonable temporal and spatial coverage in the observational record.

3.3 Observation-based data

The observational data sets for temperature and precipitation both come from the Climate Research Unit of the University of East

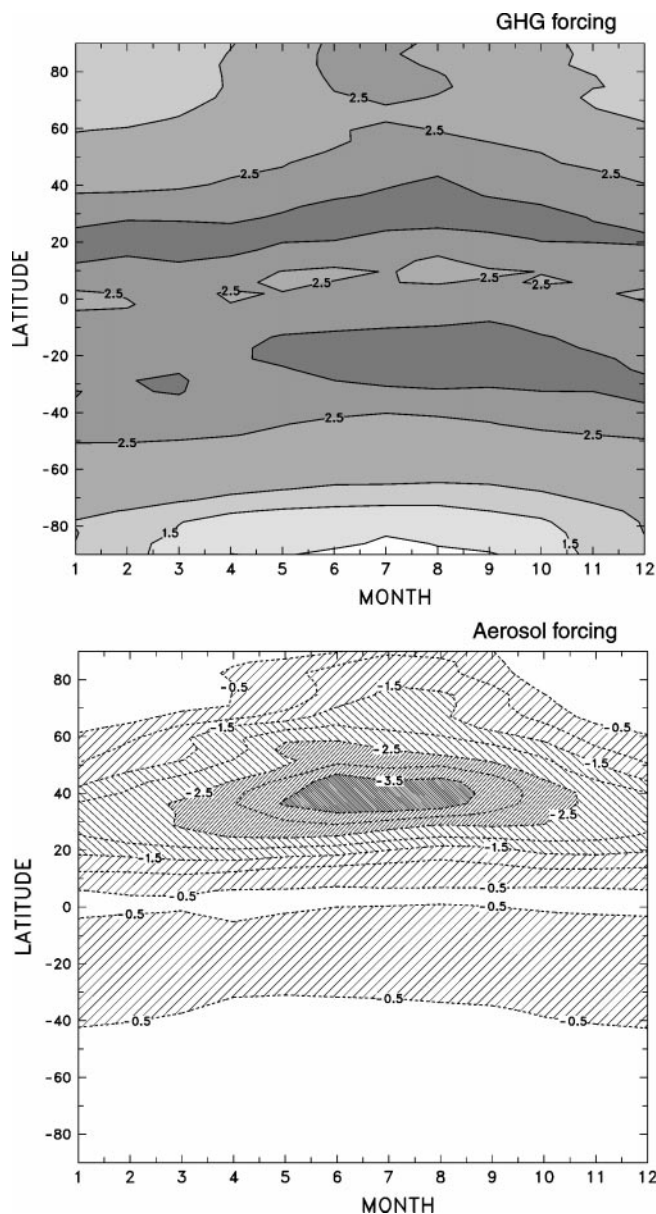


Fig. 3 The annual cycle of global mean direct *GHG* and aerosol forcing appropriate to the year 2000. Units Wm^{-2}

Anglia. The observation-based climate record for temperature is obtained as a combination of the land surface air temperature of Jones (1994) and the sea surface temperatures of Parker et al. (1995) as discussed in IPCC 1995 (Nichols et al. 1996). The data is analyzed for the 1900–1995 period and is available on a $5^\circ \times 5^\circ$ grid as monthly anomalies from the 30-year climate mean for 1961–1990. Missing data are common for much of the period at southern high latitudes and, to a lesser extent, at northern high latitudes. There are other data gaps which, as would be expected, shrink in area as time approaches the present. Temperature is the historical climate variable with the best coverage and availability and comparisons are confined largely to this variable.

Despite the considerable uncertainty in precipitation estimates, the variable’s practical importance for climate change demands attention. An attempt is made to compare simulation results with the observed historical monthly precipitation dataset for global land areas for 1900–1995. The data is available on a $2.5^\circ \times 3.75^\circ$ grid. It is an outgrowth of the climatology discussed in Hulme (1992, 1994). Some indication of data availability for both tem-

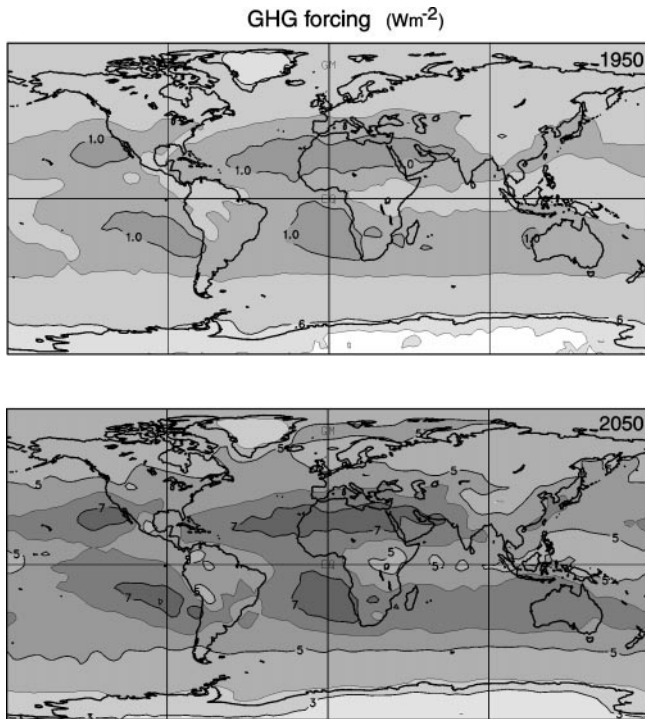


Fig. 4 The geographical distribution of annual average direct GHG forcing appropriate to 1950 and 2050. Units Wm^{-2}

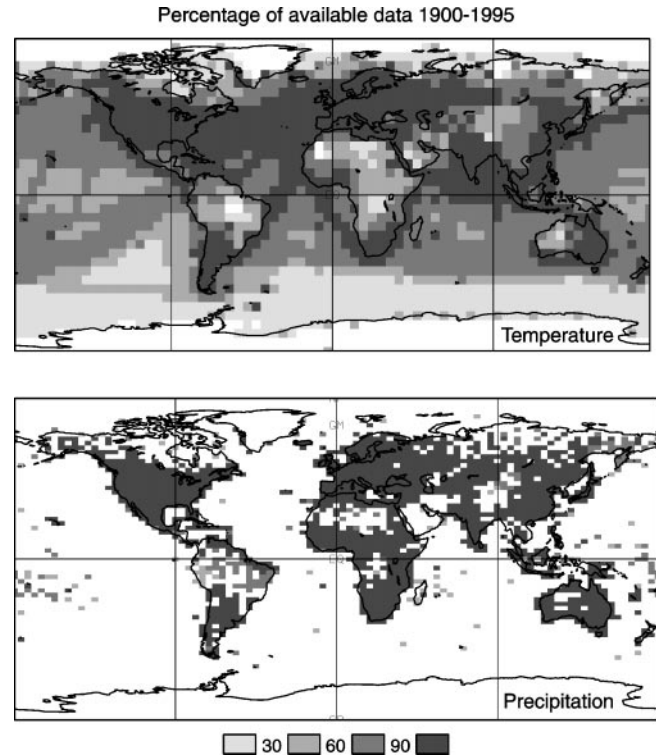


Fig. 6 The percentage of months for which observation-based temperature and precipitation data are available

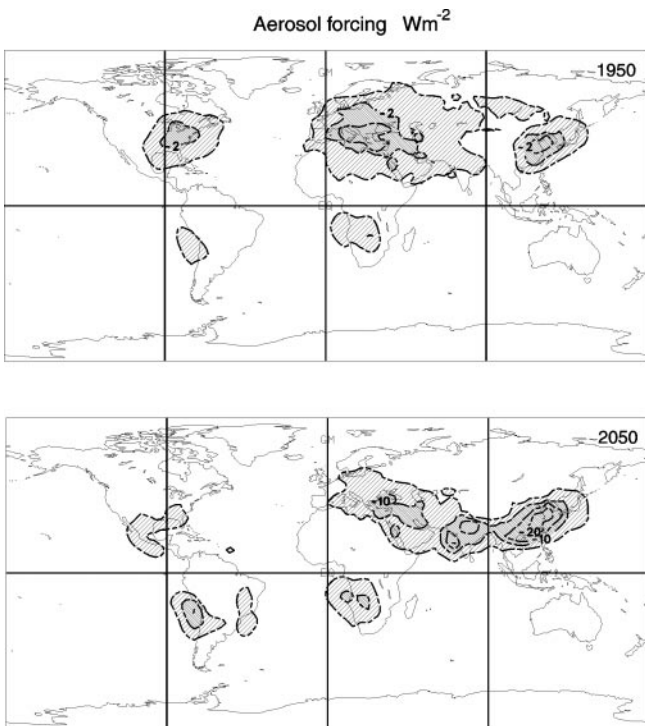


Fig. 5 The geographical distribution of annual average direct aerosol forcing appropriate to 1950 and 2050. Units Wm^{-2}

perature and precipitation is given in Fig. 6. Note that when observed data is compared to model results, the missing data mask from the observations is applied also to the model data so as to make the comparison direct.

3.4 Simple statistics

The observed and simulated data are analyzed and compared using basic statistics and basic statistical tests. Statistics of interest include average rates of change or trends in the data, means and differences of means, and variances and ratios of variances. Time series of temperature, precipitation, and other variables are considered at a point or averaged over some region. A measure of the average rate of change $b_X = dX/dt$ for period 1900–1995 is obtained as the slope of the least squares straight line fit to the data. A version of the t -test determines if the value of b_X can be statistically distinguished from zero, that is, if there is a significant trend in the data for the period. A second t -test determines if the average rates of change from different data sets may be statistically distinguished one from another, that is, if b_X is statistically different from b_Y .

The intent of the statistical tests used here is to provide context for the changes in observed and simulated climate statistics without demanding complete statistical purity. There are several related aspects that may be considered for trends in temperature for instance. It is natural to test if trends in temperature are significantly different from zero and this is done using the t -test at some level of significance. A second aspect is to test if the non-zero trends in the several data sets differ, i.e. if the temperature trends in the observations, GHG + aerosol simulations, and GHG-only simulations are significantly different from one another according to the t -test.

Two aspects of this sort of statistical testing of climate data should be mentioned. The first is the application of the standard t -test to non-independent data, i.e. data with serial correlation such as temperature. As discussed in the Appendix, the presence of serial correlation can affect the result and a modification of the test that corrects for this in an approximate way is used here. In practice, for the cases considered, the results are largely insensitive to this modification.

A second aspect arises from the nature of statistical testing. The typical statistical test may show that trends are different from one another (i.e. rejection of the null hypothesis that they are equal) but not that they are the same. Nevertheless, where there is a physical

reason (i.e. forced climate change) for the existence and magnitude of the temperature trends in the observations and simulations, the trends may be said to be “not inconsistent” with one another or with the idea of forced climate change if the test is unable to clearly distinguish between them, given the available data.

4 Surface air temperature

The global mean, zonally averaged, and local temperature evolution is investigated by comparing simulated and observed values. There are three independent realizations of the climate evolution forced with the GHG + aerosol forcing and this serves to help isolate the climate signal. It is assumed that the climate system will respond to the specified forcing scenarios with a deterministic forced signal on which is superimposed non-deterministic natural variability. Symbolically $X(t) = X_o + X_f(t) + X'(t)$ where X_o is the long term climate mean (the average of the control simulation for the model), X_f the forced climate change signal, and X' the natural variability. All quantities are implicit functions also of latitude, longitude, and height or depth. For an ensemble average (indicated by braces) over a sufficiently large number of realizations, $\{X\} \approx X_o + X_f$ and $\{X'\} \approx 0$. With three realizations, the average $\{X\}$ only approximates this. Since the deterministic forcing change varies smoothly in time, as shown in Fig. 2, it may be acceptable to fit a smooth curve to $\{X\}(t)$ to estimate $X_f(t)$. For symbolic ease, and when suitable in context, the symbols X_{ob}, X_c, X, X_g are used to denote the value of a quantity obtained from “observations”, the control simulation, one or more of the GHG + aerosol simulations (denoted without subscript), and the GHG-only simulation respectively.

4.1 Global mean temperature

Figure 7 displays the observation-based global average (indicated by angular brackets) annual mean surface air temperature $\langle T_{ob} \rangle$ together with $\langle T \rangle$ for the three individual GHG + aerosol simulations. The curves are plotted as deviations from their respective 1961–1990 averages. The observation-based curve represents a single realization of the evolution of the actual system under the forcing due to GHGs, aerosols and any other forcing of the system that may exist. The GHG + aerosol curves represent independent realizations with the same specified GHG + aerosol forcing change. The curves display a long term increasing trend together with shorter term variability and are visually similar, except for the warm period in the observed record during the 1940s and 1960s. Mitchell et al. (1995) discuss similar results for their simulation and speculate that a combination of natural fluctuations and small errors in forcing could be the reason for this apparent discrepancy.

Figure 8 displays the global mean temperature evolution for the control simulation, the observations, the average of the three GHG + aerosol simulations, and

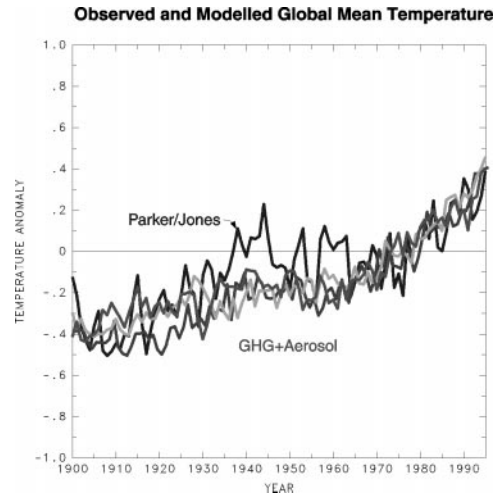


Fig. 7 Observed and modelled global annual mean temperature anomalies from the 1961–1990 climatological average. Three independent GHG + aerosol simulations are shown. Units °C

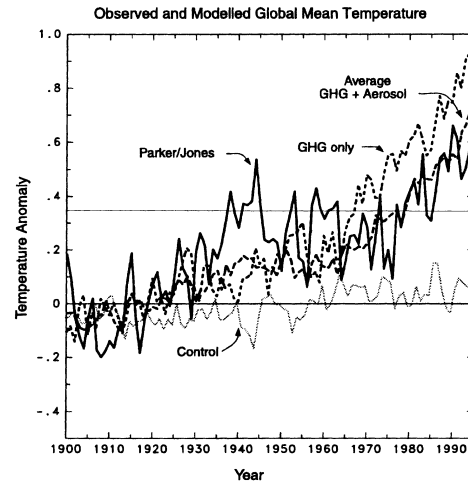


Fig. 8 Global annual mean temperature anomalies from the 1900–1930 climatological average for the observations, the average of the three GHG + aerosol simulations, and the GHG-only simulation. The thin line is three times the standard deviation from the unforced 200 year control run. Units °C

the GHG-only simulation. The curves are plotted about their 1901–1930 means. An increase of temperature of the order of 0.6 °C during the period is a feature of all of the curves except that for the control simulation. For the simulations, these trends are due to specified changes in forcing associated with changing GHG concentrations and aerosol loadings. The GHG + aerosol curve of global mean surface air temperature is visually closer to the observed curve than is the GHG-only curve which is beginning to separate from the other two toward the end of the period. There is a weak temperature trend in the control simulation, but it is small compared to that of the remaining curves.

An estimate of the standard deviation of the global annual mean temperature based on the 200 years of the

control simulation gives $S_T = 0.12$ °C so that the global annual mean temperature is expected to exceed $3S_T = 0.37$ °C less than 1% of the time provided the temperature is normally distributed. Figure 8 shows that the observation-based curve exceeds this limit (indicated by the thin horizontal line) for a few years in the warm 1940–60 period but consistently for the decade at the end of the period. The simulation for the average GHG + aerosol case (and for the individual realizations) exceeds this limit at about the same time, while the GHG-only case crosses the boundary at least a decade sooner. In the case of the model at least, the warm temperatures seen at the end of the period would be unlikely to occur by chance based on the variability of the control simulation.

If the model value of S_T were the same as that of the climate system then Fig. 8 would be good evidence of forced climate change in the real system also. As it is, and as implied by Fig. 7, the observation based record appears to exhibit somewhat larger variability than does the simulated record, although this may be the consequence of data difficulties or forcing by other than GHGs and aerosols.

4.2 Global mean temperature trends

Global mean temperature trends for the 1900–1995 period are estimated by fitting least squares straight lines to observation-based and model data. The resulting slope b_T of the fitted straight line gives the average rate of change of temperature for the period and is measured in degrees per century. Table 1 gives the standard deviation of the global mean temperature S_T , the standard deviation associated with the linear trend S_L , and the standard deviation associated with the scatter about the trend line S_e where $S_T^2 = S_L^2 + S_e^2$. The values of S_T range from 0.190 to 0.224 °C with the three GHG + aerosol values bracketing the observation-based value. The GHG-only value is the outlier at 0.276 °C. Relationships are similar among the values of S_L . The scatter, S_e , about the linear trend is largest for the observations.

The temperature trend of 0.60 °C/century in the observations is bracketed by the three GHG + aerosol simulations and close to the 0.66 °C value for the average GHG + aerosol case while the GHG-only value is larger at 0.91 °C. These trends are significantly different

Table 1 Linear trend in global mean temperature for 1900–1995 and the standard deviations associated with the temperature (S_T), the linear trend (S_L), and the scatter about the trend (S_e)

	<i>Trend</i> (°C/100 y)	S_T (°C)	S_L (°C)	S_e (°C)
Observations	0.602	0.210	0.167	0.128
Average GHG + aerosol	0.664	0.200	0.184	0.079
GHG only	0.914	0.276	0.253	0.112
GHG + aerosol1	0.595	0.190	0.165	0.095
GHG + aerosol2	0.660	0.203	0.183	0.090
GHG + aerosol3	0.738	0.224	0.205	0.091

from zero based on the t -test (which includes the modification for serial correlation). Table 2 indicates that trends differ between the control simulation and the observations and the forced simulations. The test is not able to distinguish, however, between the observed trend and the individual or average GHG + aerosol trends. The test does, however, distinguish between the GHG-only trend and the rest.

These results reiterate that the GHG-only forcing results in too rapid a temperature increase compared to observations while the inclusion of a representation of the direct effect of aerosols results in less disagreement between the modelled and observed temperature trends.

4.3 Comparison with other simulations

Table 3 summarizes global mean temperature changes from 1900 to the present simulated by other models with similar forcing scenarios (Mitchell et al. 1995; Cubash et al. 1997; Haywood et al. 1997). The values have been read from available diagrams so are uncertain to that extent. One measure of the “climate sensitivity” of a model is its equilibrium temperate change for a doubling of CO₂ (usually obtained from a simulation with a “mixed-layer” ocean component). The models in Table 3 have reported climate sensitivities ranging from a low of 2.5 °C for the Hadley Centre model to a high of 3.7 °C for the GFDL model. IPCC 1990 estimates that the climate sensitivity falls in the range 1.5 to 4.5 °C so that the CCCma model sensitivity at 3.5 °C is in the upper half of this range. In fact, two models have sensitivities near 2.5 °C while the other two have sensitivities about a degree higher.

Table 2 Statistical tests of the trends in global mean temperature. Is the t -test able to distinguish between trend values at the 5% confidence level?

Trend in	Significantly different from trend in		
	Control	Observation	Average GHG + aerosol
Observations	yes	–	–
Average GHG + aerosol	yes	no	–
GHG only	yes	yes	yes

Table 3 Climate sensitivity and simulated temperature change (°C)

Model	$2 \times \text{CO}_2$	GHG + aerosol	GHG
	Equilibrium “sensitivity”	1900–present	1900–present
CCCma	3.5	0.6	0.8
GFDL	3.7	0.7	1.2
MPI	2.6	0.5	0.8
UKMO	2.5	0.5	0.9
Mean	3.1	0.6	0.9
Standard deviation	0.5	0.1	0.2

Despite these differences in sensitivity, the agreement between observed and modelled warming trends is notable for all of the models. This suggests that it is not yet possible to distinguish between simulated global mean temperature results from models with different climate sensitivities by comparison with observations. The results in the table do suggest that it is possible to distinguish between simulations with and without aerosol effects, in other words, the difference between GHG and GHG + aerosol forcing has a greater effect on the simulated climate for the historical period than does a 1 °C difference in model climate sensitivity. This in turn suggests that it is difficult, at this time, to limit model sensitivity based on observed temperature trends.

This may be partially understood by recourse to a simple one-dimensional “global mean climate model” based on the thermodynamic equation $C_* \partial T / \partial t = -aT + (f_g - f_a)t$ where a is intended to represent the strength of climate “feedback” processes (both positive and negative) in the system, C_* is an effective heat capacity, f_g is the rate of change of GHG forcing and f_a that of the negative forcing due to aerosols, both assumed to increase linearly. T is the change in temperature about the preindustrial climate mean. The temperature response for this simple system depends on a “climate sensitivity” s which is inversely proportional to the feedback parameter a and which is defined here as $s = C_*/a$. The temperature evolution from the initial state $T(0) = 0$ may be written explicitly as $T(t) = s(f - f_a)C_*^{-1}(t - s(1 - e^{-t/s}))$. At early times, when the temperature has changed relatively little so that the feedback term is comparatively small, the change in temperature will depend mainly on the magnitude of the forcing. This is illustrated formally by expanding the exponential and retaining terms to second order to give $T \approx (f - f_a)C_*^{-1} \left(\frac{t^2}{2} - \frac{t^3}{3s} + \dots \right) \approx (f - f_a)C_*^{-1} \frac{t^2}{2}$. The climate sensitivity of the model drops out and the temperature increase is determined by the forcing (and the effective heat capacity). The implication is that temperature change at early times does not depend (strongly) on climate sensitivity and, in so far as this simple reasoning applies, models with different sensitivities would be expected to give similar results early in the simulations, although not at later times. The effect of the aerosol forcing f_a remains, however, so that it is possible to distinguish between GHG-only and GHG + aerosol cases early in the simulation in agreement with the results in Fig. 8 and Table 3.

4.4 The latitudinal structure of temperature trend

The similarity of the simulated GHG + aerosol and observed temperature evolution extends to its zonal structure. The upper panel of Fig. 9 displays $[T_{ob}]$, the zonally averaged version of the observationally-based temperature evolution. This field is less highly averaged than the corresponding global mean and suffers considerably from missing data at higher latitudes. At least

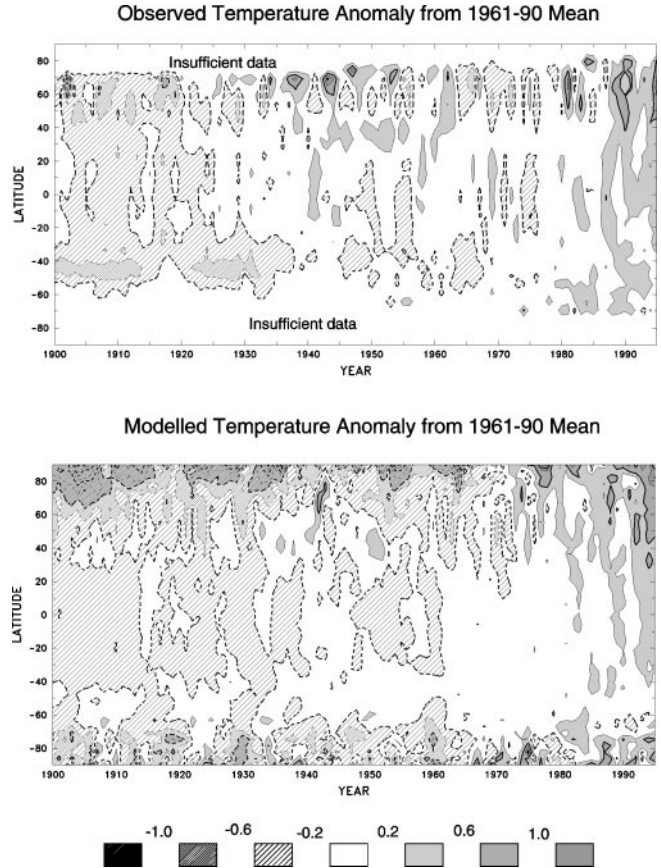


Fig. 9 Zonally averaged temperature anomalies from the 1961–1990 climatological average for the observations (*upper panel*) and from the first GHG + aerosol simulation (*lower panel*). Units °C

20% of the data must be available around a latitude circle before a zonal average is calculated. The lower panel of Fig. 9 displays the corresponding evolution of the first of the GHG + aerosol simulations. Plotted values are deviations from 1961–90 means.

The diagrams are plotted with a contour interval of 0.4 °C which is centred around 0 °C so that unshaded portions of the diagram indicate anomalies between ± 0.2 °C. Both panels of Fig. 9 show similar features: (1) there is a general increase in temperature during the period which is similar in magnitude in the two cases; (2) visually similar kinds of short term variability are superimposed on this long term trend; (3) variability and trend generally increase with latitude, especially in the Northern Hemisphere, although this is somewhat masked in the case of the observations by missing data. Results from the GHG-only simulation (not shown) have the same general features with, however, the expected larger overall trend in temperature.

4.5 Temperature trend as a function of latitude

The rate of change of temperature as a function of latitude is calculated by fitting linear trends. The results are

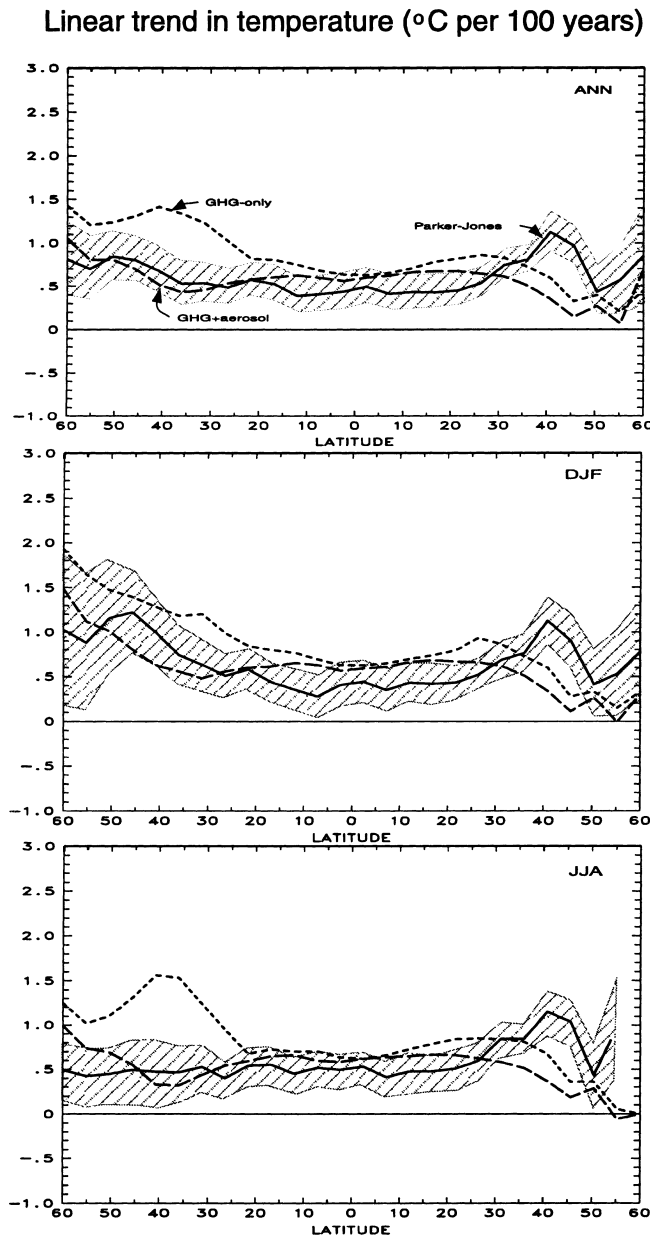


Fig. 10 Linear trend in zonally averaged temperature ($^{\circ}\text{C}/100$ years) for the annual average (ANN) and the *DJF* and *JJA* seasons from the observations, the average GHG + aerosol case, and the GHG-only case. The shaded region is the 5% confidence interval for the observations compared to the simulated trends for the GHG + aerosol case as discussed in the Appendix

displayed in Fig. 10 for the annual average and the DJF and JJA seasons. The model data is masked out in the same fashion as the observational data and, because of data availability with latitude, only the values between 60°N and 60°S are plotted. The diagrams displays the trend estimate from the observational data and a shaded confidence band together with the temperature trends for the average of the three GHG + aerosol simulations, and the GHG-only simulation. The top panel indicates an observed annual average trend (in $^{\circ}\text{C}$ per 100

years) of between about 0.4°C in the tropics to something over 1.0°C near 40°S . According to the discussion in the Appendix, the observed trend is judged everywhere significantly non-zero since the shaded band is everywhere above the zero line. The simulated trends are also significantly different from zero except for the region around 50°S .

The trends simulated with GHG + aerosol forcing fall within the shaded confidence band at most latitudes, indicating that they do not differ significantly from the observed trends. The notable exception is at southern latitudes where the observationally based trend has its maximum but where the simulated trend is small. This appears to be a clear mismatch between the observed and simulated temperature change.

The GHG-only trends fall outside the shaded band over much of the latitude range indicating that they differ significantly from the observed values. Although the GHG-only trends are generally greater than those based on observations, they are also significantly smaller than the observed trend in the southern middle latitudes, again indicating notably different behaviour between simulated and observed trends there.

Seasonal trends for DJF and JJA show many of the same features; the observed trend is generally significantly different from zero but not from the GHG + aerosol trend. The GHG-only value is generally the outlier but this is less obvious in DJF because the confidence interval has widened. The DJF and JJA trends do not differ much over most of the tropics and the Southern Hemisphere while there is a marked seasonal difference in middle and higher northern latitudes where trends are considerably larger in winter.

To summarize, a modified *t*-test is unable to distinguish between observed trends in zonally averaged temperature and those simulated with GHG + aerosol forcing, with the notable exception of the region near 40°S . The test is generally able to distinguish between observed trends and those simulated with GHG-only forcing. Seasonal effects in the rate of warming are evident primarily at middle and higher northern latitudes where warming is strongest in winter.

4.6 The geographical distribution of temperature change

The potential ability of climate models to simulate the geographical distribution of climate variables and their changes is one of their major strengths. It becomes more difficult, however, to discern a forced climate signal in the presence of natural variability as scale and averaging decrease. Two approaches to analyzing and comparing the observed and simulated pattern of temperature change are attempted. The first compares the patterns of differences in 20-year mean temperatures from the beginning and end of the observational record, and the second compares the geographical distribution of linear trends in local temperatures.

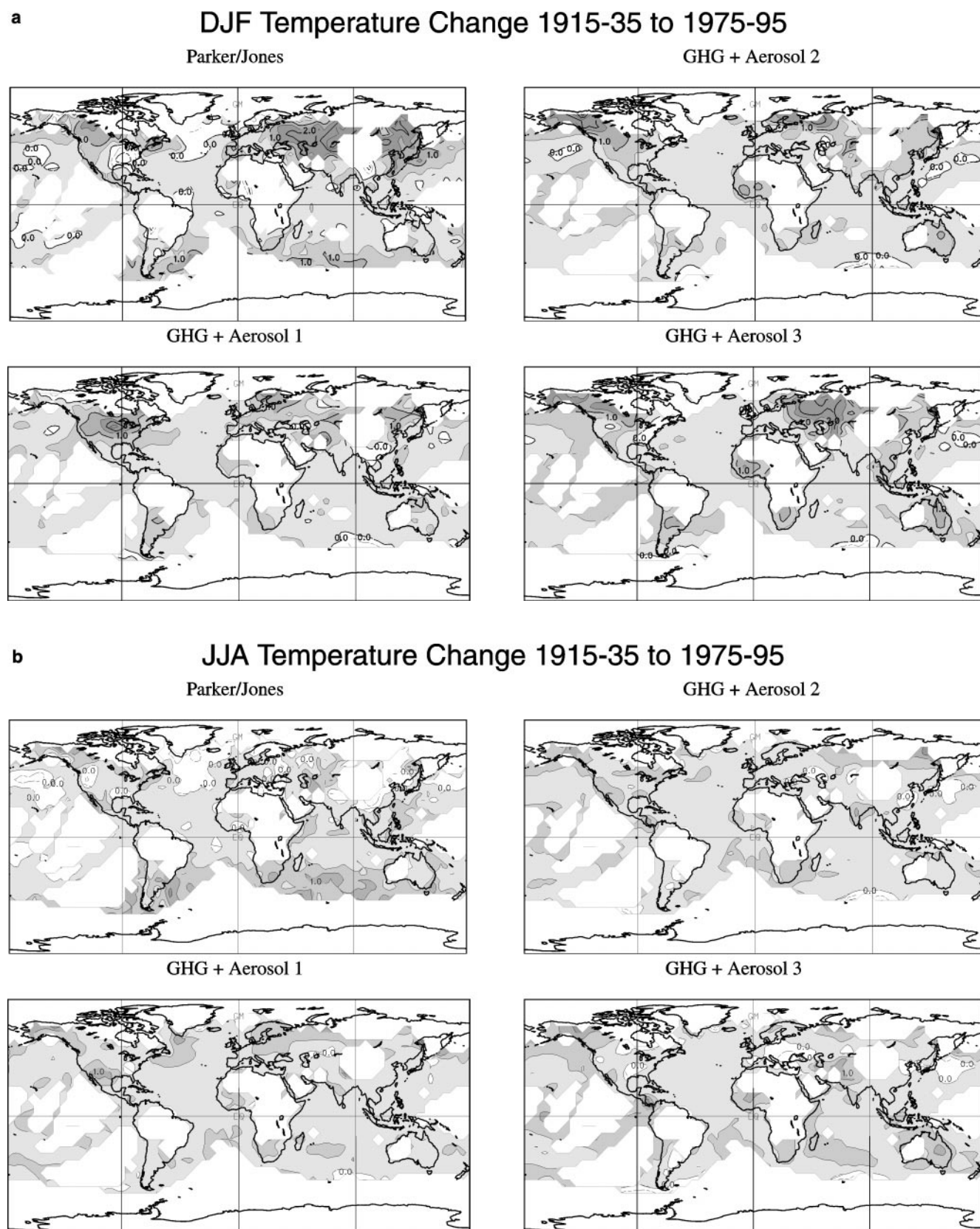


Fig. 11a,b The geographical distribution of the difference in the 20-year average temperatures between 1915–35 and 1975–95 for the DJF and JJA seasons. Both the observationally based value and those from

the three independent GHG + aerosol simulations are shown. Model data is masked out with the observational missing data mask. Units °C

Figure 11 shows the geographical distribution of the change in the 20-year average temperature between 1915–35 and 1975–95. The difference between the two periods is, symbolically, $\delta T = \delta T_f + \delta T'$ where the T here represent 20 year averages. The result is the difference in the forced signal together with the difference in a particular realization of the natural variability of the system. Since the forced signal is growing with time, its difference δT_f will increase as the time interval between 20-year periods increases while the magnitude of the natural variability $\delta T'$ presumably will not. This is the reason for choosing the difference between a 20-year period early in the century and one near the present (an even earlier 20-year period could have been chosen but the observed data is even sparser). The temperatures are not plotted if there is less than 20% of the data available in either of the two periods.

Figure 11 gives the DJF and JJA seasonal average temperature difference δT_{ob} between the 1915–35 and 1975–95 averages based on the observed data set, together with δT from each of the three independent GHG + aerosol simulations. The plots have a similar appearance and there is nothing to immediately indicate which panels are from the model and which is based on the observations. In particular, the figure shows: (1) a general overall warming during the period, (2) the largest values of the warming in the Northern Hemisphere over the central and northern areas of the continents; (3) larger warming over northern continental areas in DJF, (4) Southern Hemisphere warming largest for mid-latitude oceans; (5) cooling in parts of the north Pacific and Atlantic and, in the model, in the Southern Ocean (largely masked out here); and (6) patchy cooling over parts of continents, e.g. north of the Mediterranean and eastern North America especially in JJA. The difference between the several realizations gives an indication of the unforced natural variability in the model.

Although there is certainly a “family resemblance” between the various panels in Fig. 11, this is at best qualitative support for agreement between simulated and observed temperature changes. The mean square difference $\langle d^2 \rangle$, where $d = \delta T - \delta T_{ob}$ and angular bracket indicates a (suitably masked) global area average, gives a qualitative measure of (dis)agreement. A scaled version $f = \langle d^2 \rangle / (S_m^2 + S_{ob}^2) = 1 - \beta r = 1 - p$ of the mean square difference ranges from 0 to 1. It is related to the pattern correlation r where S^2 is the spatial variance of the modelled or observed field and

$\beta = 2S_m S_{ob} / (S_m^2 + S_{ob}^2)$ is a slowly varying function of the spatial variances of the two fields which is 1 if the variances are the same. Here f , the measure of disagreement between the two fields is complementary to p the measure of agreement closely related to the usual spatial correlation.

Table 4 gives numerical values of the spatial correlation p and the first row compares observed with simulated temperature change patterns with values with ranging from 0.49 to 0.62 (indicating explained variance of from about 25 to 40%). There is no clear distinction between the GHG + aerosol and GHG-only cases. The simulation results are generally better correlated with one another than with the observations (although the correlation between simulated values falls as low as 0.62 in one case).

While the table gives a numerical measure of the agreement between the observed and simulated patterns, the spatial correlation of the fields means that a statistical significance measure cannot be readily supplied. Nevertheless, only about 15 degrees of freedom would be necessary to declare the correlation coefficient to be different from zero at the 5% level. While there is some coherence between simulated and observed geographical patterns of temperature change it is modest and, unlike the case for the global mean or the zonal mean temperature, the GHG-only result does not clearly stand apart.

4.7 The geographical distribution of temperature trend

A second “differential” approach concentrates on the geographic pattern of linear trend calculated for each point. The observation-based and simulated trends for the three GHG + aerosol cases are shown in Fig. 12. There are, as expected, generally positive trends, although there are patches of negative or cooling trends in both observations and model results.

Figure 13 indicates those points for which the simulated annual average trend of temperature is judged to be statistically different from the observed trend at the 5% level using the modified t -test discussed in the Appendix. Table 5 gives the percent of the (non-masked out) area where these limits are exceeded. The average GHG + aerosol simulation has roughly the same fraction of area with trends judged to be significantly above

Table 4 Spatial correlation between 1915–35 to 1975–1995 temperature difference patterns (see text)

	Observations			Average GHG + aerosol			GHG only			GHG + aerosol1		
	Ann	DJF	JJA	Ann	DJF	JJA	Ann	DJF	JJA	Ann	DJF	JJA
Average GHG + aerosol	0.49	0.55	0.52									
GHG only	0.54	0.62	0.47	0.80	0.74	0.68						
GHG + aerosol1	0.47	0.49	0.49	0.97	0.90	0.93	0.81	0.72	0.70			
GHG + aerosol2	0.47	0.48	0.46	0.99	0.94	0.95	0.79	0.70	0.65	0.94	0.77	0.84
GHG + aerosol3	0.50	0.58	0.51	0.97	0.93	0.93	0.77	0.70	0.62	0.90	0.74	0.79

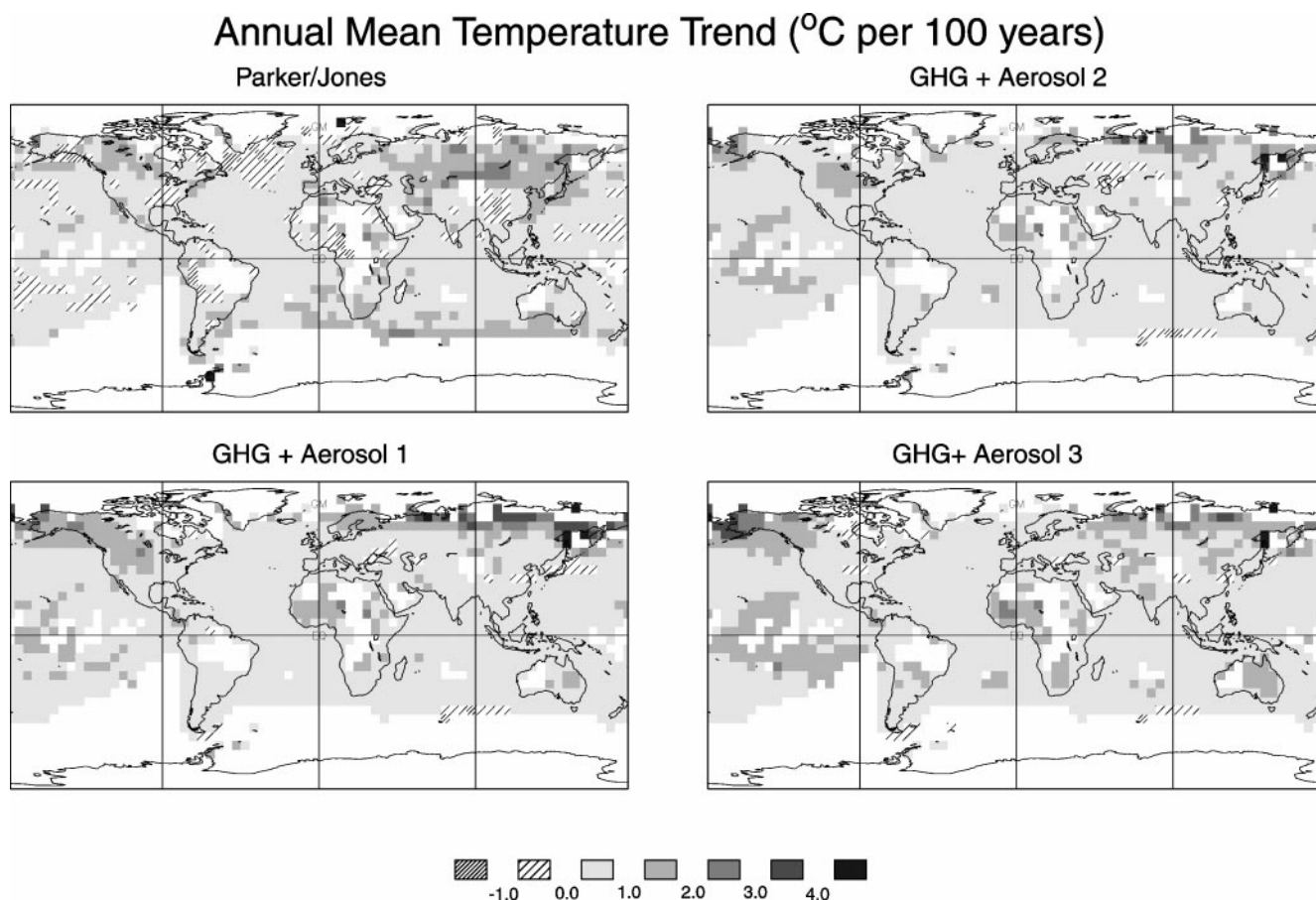


Fig. 12 The geographical distribution of linear temperature trend for the 1900–1995 period for the annual average. Both the observationally based value and those from the three independent GHG + aerosol simulations are shown. Units °C per 100 years

and below the observed temperature trend. This contrasts with the GHG-only case where more than twice the area exhibits significantly different simulated temperature trends and where, moreover, the trends are predominantly larger than those of the observations. Finally, the control simulation exhibits the reverse relationship, with only a few points exhibiting trends significantly larger and many more exhibiting trends significantly smaller than the observation-based trends. This is all consistent with the physical picture of a forced climate change with trend in the observations which agrees reasonably well with (or at least is not trivially distinguishable from) the trend simulated in the GHG + aerosol forced case while the trend simulated in the GHG-only case is too great.

4.8 Regional temperature change

The IPCC 1990 and 1995 scientific assessments focus attention also on the land areas of five sub-continental scale regions: Central North America (35–50°N, 80–105°W), South East Asia (5–30°N, 70–105°E), Sahel (10–20°N, 20°W–40°E), Southern Europe (35–50°N, 10°W–

45°E), and Australia (10–45°S, 110–155°E). Regional climate change, in this context, represents averages over regions which are of scale intermediate between the latitude band of a zonal average and local values on the grid. Flato et al. (2000) give some attention to these regions for the control simulation. Various model studies and intercomparisons based on these regions are given in the IPCC reports and associated studies such as that of Kittel et al. (1998) which considers the regional climate response at approximately the time of CO₂ doubling in simulations forced with 1% per year increasing CO₂. Here, by contrast, the simulated and observed climate change for the period 1900–1995 is compared.

Figure 14 displays temperature anomalies from the 1901–1930 average, smoothed with a 10-year running mean, for the land portions of the five regions for the

Table 5 Percentage of area for which the simulated trend in temperature is significantly greater (first entry) or less (second entry) than the observed trend at the 5% level

	Annual		DJF		JJA	
Average GHG + aerosol	18	12	14	12	14	9
GHG only	35	4	22	6	26	4
GHG + aerosol1	14	12	9	10	8	9
GHG + aerosol2	16	11	13	11	12	9
GHG + aerosol3	19	9	15	10	15	7
Control	3	39	2	29	3	30

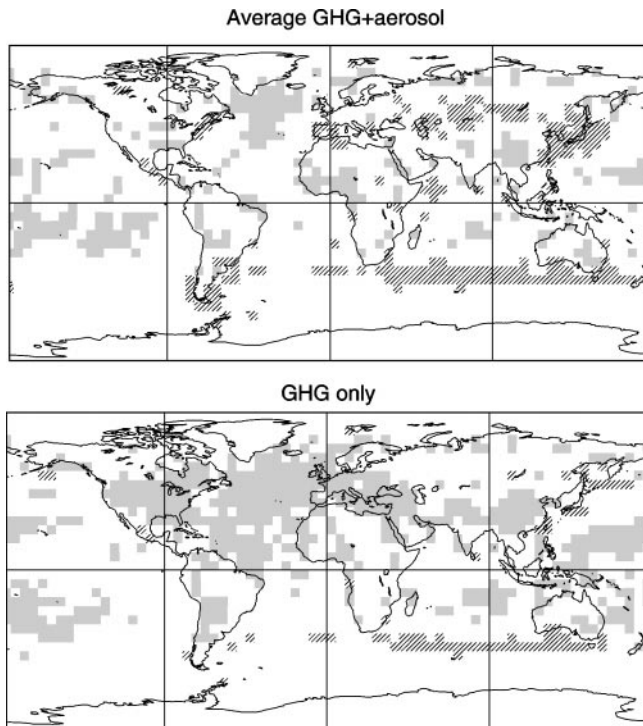


Fig. 13 Those regions where the approximate 5% significance band for the observation-based linear trend of annual average temperature for the 1990–1995 period is exceeded above (shading) or below (hatching) by the corresponding linear trend in the average of the GHG + aerosol simulations and the GHG-only simulation

observations, the average of the GHG + aerosol simulations, and the GHG-only simulation. The bottom right hand panels of the figures summarize the results of regional trend analysis where the box outlines the 5% significance band about the observed trend as described in the Appendix.

For the North American region, the temperature pulse in the 1930–40s is a notable feature together with a subsequent cooling followed by recent warming. Interestingly, the average GHG + aerosol simulation shows a somewhat similar although distorted minimum toward the end of the period before a strong rise begins. This is in contrast to the GHG only simulation which shows comparatively strong warming over the latter half of the period. According to the bottom right hand panel of the figures, the GHG-only trend is significantly greater than that of the observations in both the annual and JJA cases while the GHG + aerosol trend is within the significance band in all seasons.

South East Asia displays modest temperature increases where simulated changes tend to be somewhat greater than those observed. There is not much separation between the GHG + aerosol and GHG only cases here and trends of this magnitude fall outside the confidence band except in DJF.

In the Sahel region the simulated warmings are significantly larger than those observed for all seasons. The Sahel is a comparatively small region located where

there are relatively strong north-south gradients in climate parameters. This makes it difficult to simulate the current climate successfully and makes the regional average sensitive to climate shifts and changes.

For Southern Europe, GHG + aerosol and observed temperature evolutions are reasonably similar while the GHG only simulation shows larger temperature increases. The GHG + aerosol trends cannot be distinguished from those of the observations while that of the GHG-only simulation can be distinguished in JJA and on the annual average.

Finally, the GHG + aerosol and observed temperature evolution for the Australia region is similar and their linear trends cannot be distinguished. The trend simulated in the GHG-only case is marginally different for DJF and annual average temperatures.

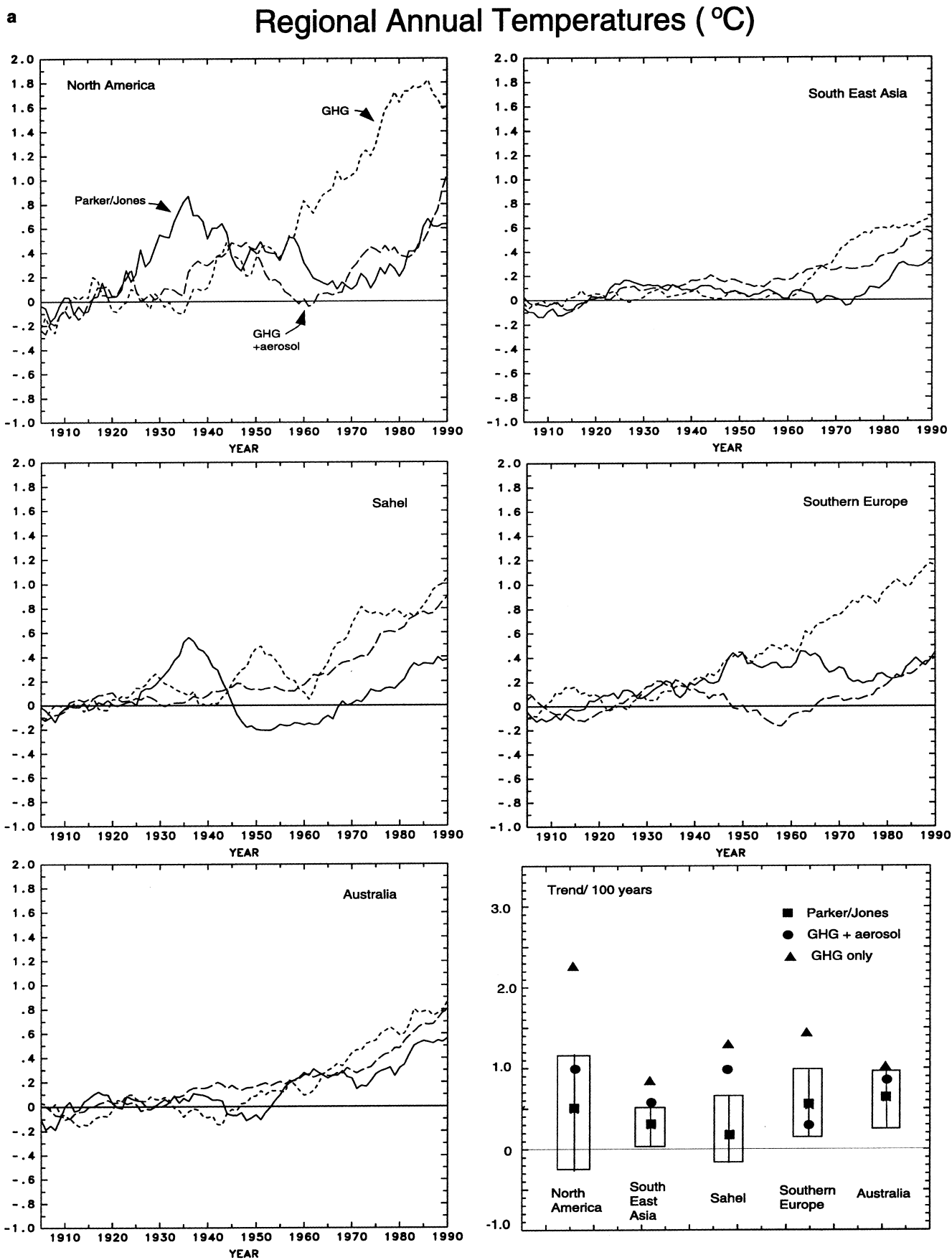
These results suggest some modest regional agreement between observed temperature evolution and that simulated with GHG + aerosol forcing. There is less agreement between the observations and the GHG-only simulation. There is disagreement between observed and simulated temperature evolution for the Sahel.

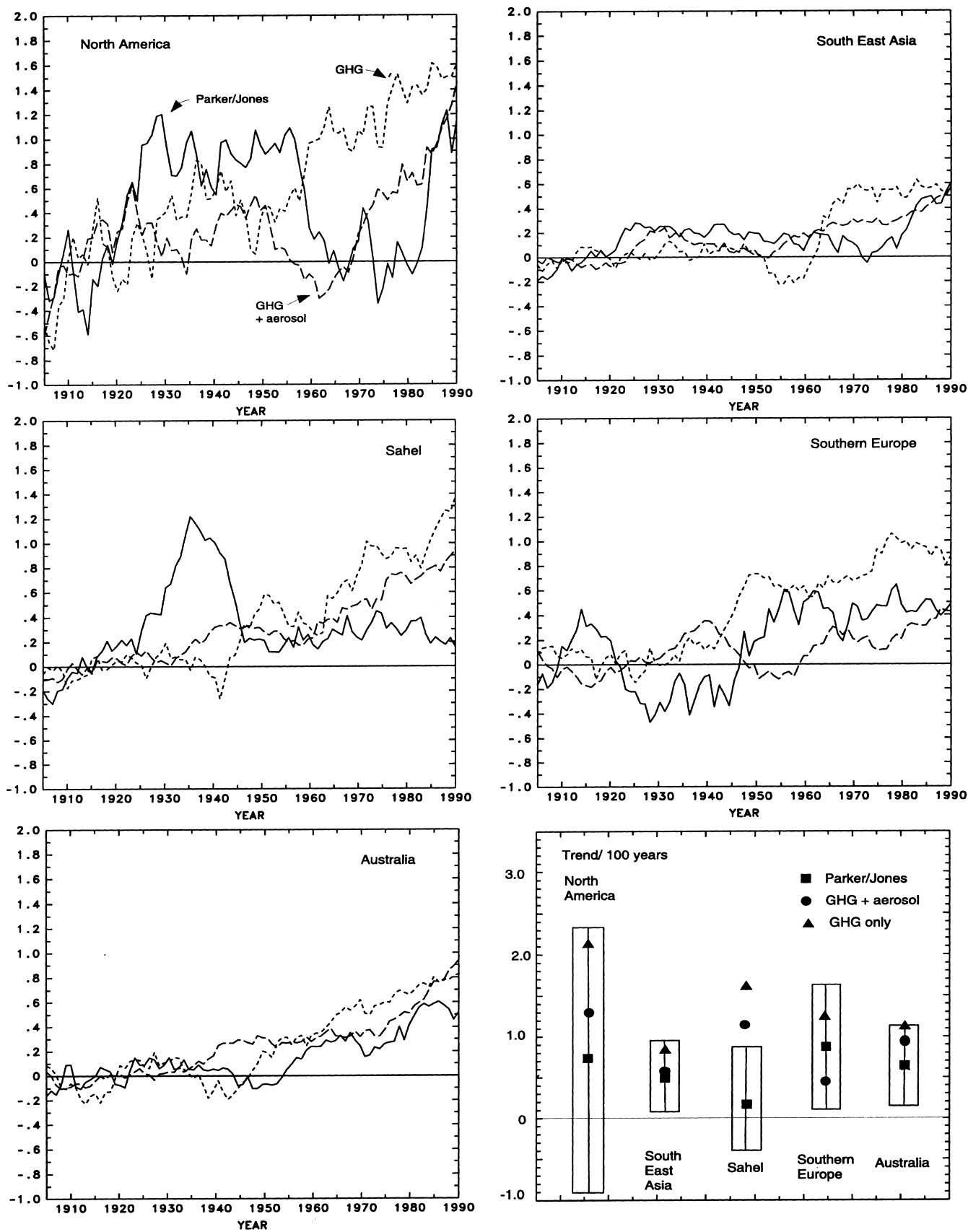
5 Precipitation

Temperature and precipitation are both critically important climate variables for practical economic, social, and environmental reasons but they have very different properties. Temperature is relatively easy to observe, has comparatively large and stable structures, and is directly connected to the perturbation in the energy budget implied by the GHG and aerosol forcing. Precipitation, on the other hand, has comparatively small-scale structures and large variability. The observed precipitation record is confined largely to the land. Nevertheless, because of the immense practical importance of perturbations to the hydrological cycle, some attempt is made to compare the twentieth century evolution of the observation-based precipitation amounts with those simulated by the model. As indicated in Fig. 3.6, the observed record is relatively complete in time where it exists over land. Information is absent for certain geographical land areas and everywhere over the oceans, with the exception of island stations.

5.1 Land precipitation change

Figure 15 displays the land average precipitation for the period 1901–1995 paralleling the curves for temperature. It is displayed as the change about the 1901–1930 mean, as a percentage of the average for the entire period, and is smoothed with a 10-year running mean. The presentation of precipitation change as a percentage is a useful scaling of this variable both locally and globally since the impact of precipitation changes depend on the climatological base level. Note that in the comparisons between observed and simulated precipitation rates, the



b Regional DJF Temperatures ($^{\circ}\text{C}$)

c

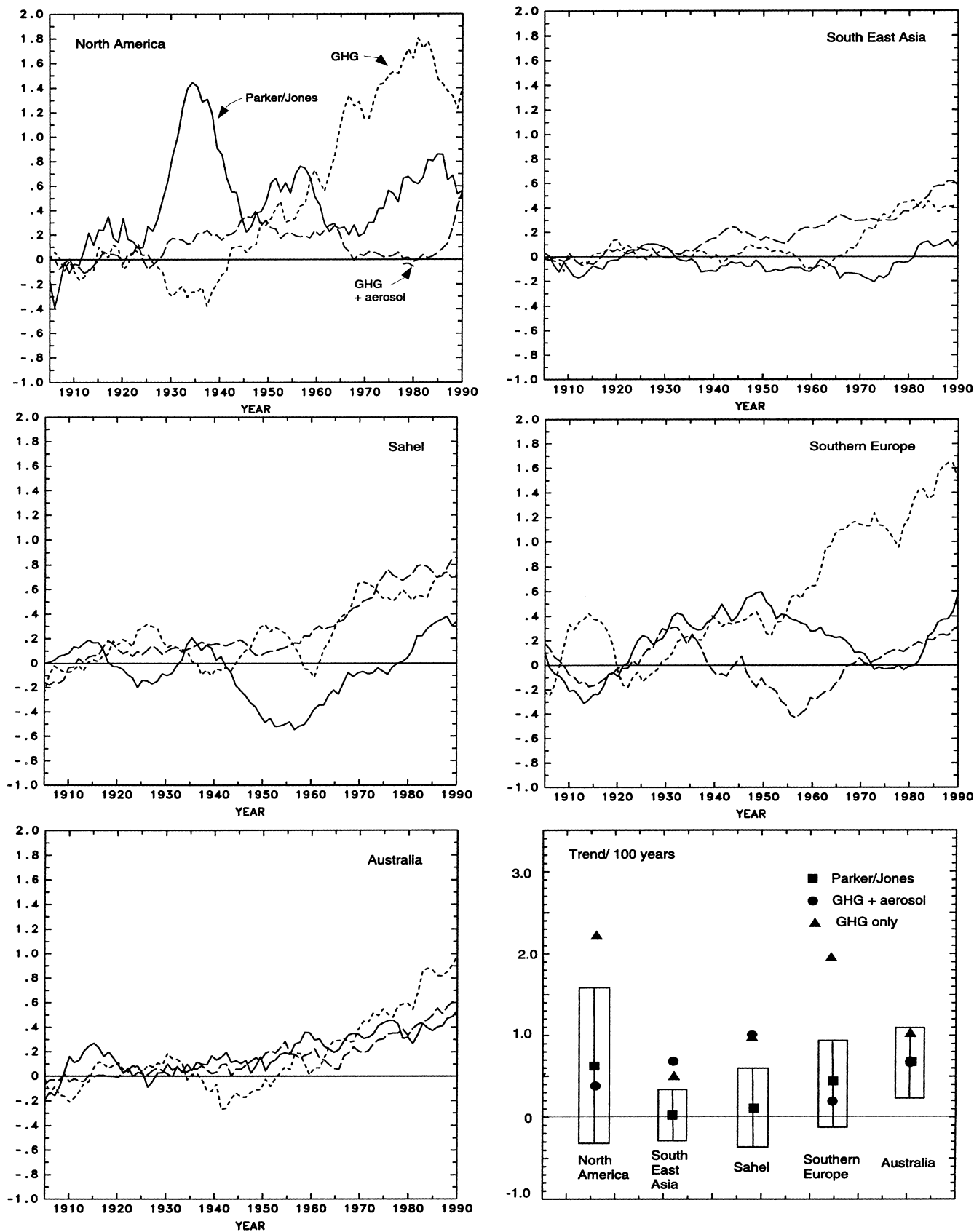
Regional JJA Temperatures ($^{\circ}\text{C}$)

Fig. 14a–c Temporal evolution and trends for the five IPCC regions. Units °C

model data has been masked out where observations are missing and that this may introduce regional biases because of the strong gradients in the precipitation field.

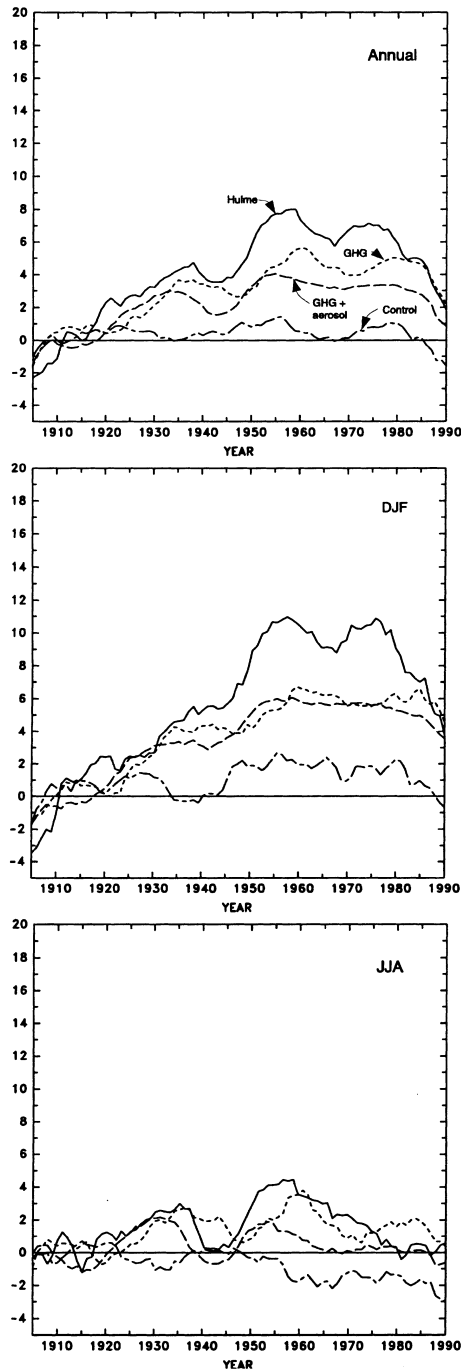


Fig. 15 The annual mean, DJF, and JJA land average precipitation rate anomalies from the 1900–1930 climatological average, expressed as a percentage and smoothed with a 10 year running mean, for the observations, the average of the three GHG + aerosol simulations, and the GHG-only simulation. The model data is masked with the missing data mask from the observations. Units percent

The calculated average land precipitation rate is characterized, in both the observations and the simulations, by an increase during the first half of the century, a subsequent plateauing and thence a decrease toward the end of the period. The linear trends in precipitation for the observations, the average of the GHG + aerosol simulations, and the GHG-only simulation are all significantly different from zero for the annual average and for the DJF season but not for the JJA season (as indicated in Table 6) but they are not significantly different from one another. On the face of it, this supports an observed and simulated increase in precipitation over land although, unlike the case for temperature, it is not possible to distinguish between the average GHG + aerosol and GHG-only cases. The precipitation for the unforced control simulation is also plotted in Fig. 15 and shows less evidence of long term change than do the observations and the forced simulations.

Such conclusions must be viewed in context, however. The observational data mask preferentially samples certain geographical regions and also changes somewhat in time. Contributions from more polar latitudes are not included in the land average and contributions from some tropical land areas of high precipitation are relatively poorly sampled, as indicated in Fig. 3.6. The results from the control simulation give some information on the effect of the sampling on the land average precipitation. The average land precipitation rate is calculated as $P_{land} = \langle mP \rangle / \langle m \rangle$ where $\langle \rangle$ represents the area average and m the data mask which is zero where there is no data and one elsewhere. For P_o, P_f, P' the long term mean, forced climate change, and natural variability as before then also $P_{land} = \langle mP_o \rangle / \langle m \rangle + \langle mP_f \rangle / \langle m \rangle + \langle mP' \rangle / \langle m \rangle$. If all data exist then $m = 1$ and the first term is a constant and is removed by calculating anomalies as is done for Fig. 15. However, this term can contribute to an apparent evolution of the land average precipitation for masked data if m varies in time. For the case of the control simulation where there is no forced component and the natural variability is suppressed by the running mean, $P_{ctl} \approx \langle mP_o \rangle / \langle m \rangle$ and the behaviour of δP_{ctl} in Fig. 15 gives some idea of how the curves might be affected by changes in sampling. The effect is small for the annual and DJF averages and the linear trend in the control simulation precipitation can not be distinguished from zero. For JJA, however, the linear trend in the control simulation is negative and significantly different from zero so sampling is presumably at least partially responsible for the tendency of all curves to decrease in the latter part of the period in JJA. Despite this sampling effect, the overall result is of a general increase in precipitation over land on the annual average and for DJF in both observations and simulations.

5.2 Regional precipitation change

While the overall land average precipitation rate may be affected by sampling, this is less the case for regional

Table 6 Significant linear trends in precipitation. A positive trend significant at the 5% level is indicated by + and at the 1% level as ++ and similarly for negative trends

	Observations			Average GHG + aerosol			GHG only		
	Ann	DJF	JJA	Ann	DJF	JJA	Ann	DJF	JJA
All land	++	++	++	++	++		++	++	+
North America	+								--
SE Asia									
Sahel	--	-	--	--		--			
S Europe							-		--
Australia					-				

values as is verified by direct comparison of masked and unmasked results. Of course, the larger variability of precipitation at the regional scale makes the detection of precipitation change more difficult and there is little visual evidence of precipitation change in plots of regional precipitation amounts. Few regions have linear trends significantly different from zero for either the observations or the simulations as indicated in Table 6. In summary: (1) the evidence for regional precipitation change in the observed record is generally weak; (2) evidence for regional precipitation changes in simulation results with GHG + aerosol and the stronger GHG-only forcing is also generally weak; and (3) some modest evidence for a decreasing trend in Sahelian precipitation is seen in the observations and the GHG + aerosol simulation. The results reiterate the less robust nature of forced climate change in precipitation compared to climate variables such as temperature.

5.3 The geographical distribution of precipitation rate trend

The geographical distribution of linear trend in precipitation rate for the observations and the GHG + aerosol and GHG-only simulations is plotted in Fig. 16. The pattern of trend for the average GHG + aerosol case is smoother than that of the observations as would be expected as a consequence of the averaging the three GHG + aerosol cases together. The relatively small scales and rapid variation of the trend in observed precipitation, and the lack of observations over the oceans, makes it difficult to relate the observed and simulated quantities despite some visual connections. A relatively small fraction of points have trends which are statistically different from zero and the spatial pattern correlation between the trends in the observations and the simulations, although positive, is low (0.13). The connection, if any, between the geographical distribution of observed and simulated land precipitation changes is unclear from the available information.

6 Summary

The CCCma coupled climate model is used to perform five climate simulations with three different forcing sce-

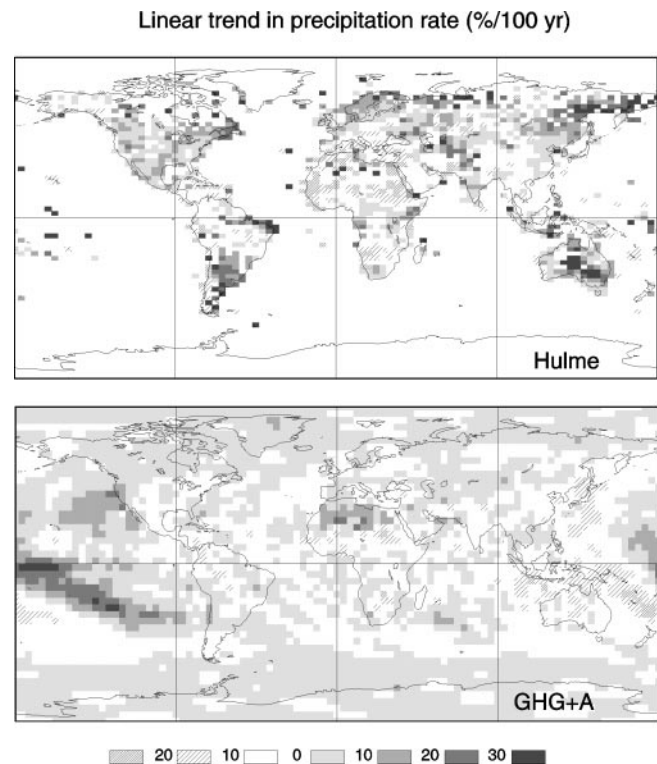


Fig. 16 The geographical distribution of linear trend in annually averaged precipitation rate, expressed as a percentage, for the observations and for the average of the three GHG + aerosol simulations. Units percent per 100 years

narios. Data for the years 1900–2100 is retained for a control simulation, three independent simulations with increasing GHG and direct sulphate aerosol forcing, and a simulation with GHG forcing only (some of this data is available for applications or other uses at <http://www.cccma.bc.ec.gc.ca>).

The forcing prescription for the simulations is described and the resulting direct radiative forcing change, as seen by the model, is calculated and displayed at 50 year intervals. The geographical distribution of the direct GHG forcing is very smooth and the pattern also scales smoothly with time. The aerosol forcing shows a much more distinct geographical pattern with maxima over industrial areas in the Northern Hemisphere.

The observation-based temperature changes for the period of the instrumental record, from 1900 to 1995,

are compared with simulated values. Observed global mean temperature changes and those simulated for GHG + aerosol forcing, show reasonable agreement for the 1900–1995 period. In particular: (1) observed and simulated global mean temperature trends are statistically significantly different from zero; (2) the GHG-only trend is significantly different (i.e. larger) than that of the observations and the GHG + aerosol cases; and (3) the statistical test cannot, however, distinguish between the temperature trends simulated in the GHG + aerosol cases and the observed trend. Similar results generally apply to the latitudinal distribution of trends in zonally averaged temperature.

Observed and simulated regional and geographically distributed temperature evolution shows similarities to global and zonally averaged evolution, although the agreement is not as good. The change in temperature from a 20-year period early in the century (1915–35) compared to a recent 20-year period (1975–95) contains the forced change over the 60-year interval as well as the unforced natural variability. Qualitative similarities are apparent in the observed and modelled geographical patterns of this change but a statistical significance measure of the correspondence is not readily available. A spatial correlation measure of the agreement of the patterns is positive, indicating some coherence between the simulated and observed geographical patterns, but the simulated patterns agree better among themselves than with the observed pattern. There is no clear distinction between GHG + aerosol and GHG-only patterns in their agreement with observations.

The geographical distribution of linear temperature trends calculated at each point for the period also suggests some agreement between simulated and observed values. The simulated trends are compared with the observed values, and regions where the simulated values fall outside of the confidence band about the observations, and hence are judged significantly different, are obtained. The local temperature trend for the average of the GHG + aerosol simulations falls equally above and below this confidence band for about 10% of the area. Simulated temperature trends exceed the upper limit in the GHG-only case about twice as much and the lower limit about half as much as for the GHG + aerosol case. The temperature trends in the control simulation, as expected for this unforced case, fall predominantly below the confidence band. All of this is consistent with the physical picture of forced temperature trend in the GHG + aerosol simulations that is reasonably similar to the observed trend and which is smaller than that for the GHG-only case.

Regional climate change in the context of a global model focuses on five regions chosen by the IPCC (IPCC 1990, 1995) as being of particular interest. Observation-based regional temperature trends are positive but are significantly different from zero for only about half of the cases considered. There is less disagreement

between observed and simulated trends for the GHG + aerosol simulations than for the GHG-only simulation.

In brief, global means and zonal averages show reasonable agreement between modelled GHG + aerosol and observed temperature changes during the current century. There is poorer agreement or disagreement between observations and the results of the GHG-only simulation. The regional and geographical patterns of temperature change also show some agreement between observations and simulations but the agreement is less direct and the distinction between the GHG + aerosol and GHG-only cases less clear.

Precipitation is a primary meteorological variable of considerable practical importance and interest. It is characterized by smaller spatial and temporal scales than is temperature and observational data for precipitation is available mainly over land. The “land average” observational data suggests an increase in precipitation during the period as does the model data (although the percentage increase is somewhat smaller) when it is treated with the same data mask as for the observations. Trends in average precipitation rate are judged to be statistically significantly different from zero for the annual average and for the DJF season, but the test does not distinguish differences between any of the observed, the GHG + aerosol, and the GHG-only trends. These results apply to the sampled region rather than to all land as there is apparently some effect due to changes in data availability over time. Regional and geographic distributions of precipitation changes do not show clear evidence of a climate change signal.

The change in climate forcing from 1900 to the present is relatively small compared to that projected from the present to the year 2100. Nevertheless, there is some correspondence between modelled and observed temperatures in terms of global and zonal means and a weaker correspondence for regional and geographical distributions for the period. The results generally support the existence of anthropogenic climate change and the importance of both GHGs and (the direct effect) of aerosols in determining the magnitude of that change. There remain many uncertainties including difficulties with the data record, uncertainties in forcing, and especially deficiencies in model response and sensitivity. However, if the projected forcing scenario from the present to the year 2100 is at all reasonable, the rate of change of climate would be expected to accelerate rapidly and the climate change signal to become increasingly prominent in the twentyfirst century.

Acknowledgements We should like to thank S. Lambert for obtaining the observational data and processing it into our standard format for use, F. Majaess for support in matters computational and data management, F. Zwiers who provided statistical advice, and our colleagues in the CCCma who contributed their interest and advice over the considerable period of these simulations. T. Delworth and R. Stouffer also provided helpful comments.

Appendix: statistical tests

A1 Visualization of results

The t -test of the difference $d = b - b_o$ between two quantities operates by asking if $t = |d|/S_d > t_c$, that is, if the t -statistic, which depends on the difference d and an estimate of its standard error S_d , exceeds some critical value t_c . b is deemed to be significantly different from b_o at some chosen confidence level if it falls outside the significance band $b_o \pm t_c S_d$. This latter relation is used provide a visual context for the comparison of observed and simulated statistics in Fig. 10 and subsequently.

In the case of Fig. 10, b is the trend for the average of the GHG + aerosol simulations, b_o is the observed trend estimate, and S_d is the associated standard error. If the GHG + aerosol curve falls outside of the shaded significance band, the observed and simulated values are deemed to be significantly different at that latitude. This applies also to the simulated trend for the GHG-only case since the estimate of the standard error, and hence the width of the significance band, is very nearly the same. As well, where the shaded significance band does not include the zero line, b_o is significantly different from zero (the appropriate standard error is smaller than S_d and the significance band narrower for the test against zero so that the test is conservative when applied in this way). The other trend values plotted in the figure are similarly different from zero if they are no closer to the zero line than half the width of the shaded confidence band. Finally, similar considerations apply to the regional confidence bands shown in Fig. 4.9.

A2 Statistical tests of non-independent data

Standard statistical tests such as the t -test assume that the data entering the calculation are statistically independent. Zwiers and von Storch (1995) (ZvS), for instance, point out that if there is serial correlation in the data, the value of S_d calculated in the usual way is generally an underestimate so that the resulting t -statistic is overestimated, the test does not perform at its nominal level of significance, and the null hypothesis is rejected more frequently than it should be. ZvS propose to modify the estimate of S_d to take correlation into account and a variant of this general approach is used to obtain approximate corrections to the quantities that enter the statistical tests used here.

For the random variable $X = \mu + x$ the statistics of interest are the sample mean $\bar{X} = \frac{1}{n} \sum_{x=1}^n X_x$ and sample variance $S^2 = \frac{1}{n-1} \sum_{x=1}^n (X_x - \bar{X})^2$ together with the difference of means $d = \bar{X}_2 - \bar{X}_1$ and the associated t -test statistic $t = d/S_d$. Here, $Ave(\bar{X}) = E(\mu + x) = \mu$, $Var(\bar{X}) = E\bar{X}^2 - \mu^2 = E(x^2) = \sigma^2$, and $Ex_x x_\beta = \sigma^2 r(\alpha - \beta)$, assuming the autocorrelation depends only on lag. If the data are independent, $r = 0$ but if serial correlation exists in the data then $r \neq 0$.

With serial correlation in the data, the sample statistics have the expected values as follows

$$Ave(\bar{X}) = \mu, \quad Var(\bar{X}) = \frac{\sigma^2}{n} (1 + C)$$

for the sample mean

$$Ave(s^2) = \sigma^2 \left(1 - \frac{C}{n-1} \right)$$

for the sample variance and, for samples of the same size

$$Ave(d) = \mu_2 - \mu_1, \quad Var(d) = \frac{\sigma_1^2 + \sigma_2^2}{n} \left(1 + \frac{\sigma_1^2 C_1 + \sigma_2^2 C_2}{\sigma_1^2 + \sigma_2^2} \right)$$

for the difference of means.

The factor $C = 2 \sum_{\alpha=1}^{n-1} (1 - \alpha/n) r(\alpha)$, depends on the serial correlation and (usually) increases the variance over what it would be if the data were independent.

Similar relations apply to the linear trend regression coefficient b for

$$X = \mu + \beta t + x \quad \text{and} \quad b = \frac{1}{Dn} \sum_{x=1}^n X_x t_x; \quad D = \frac{1}{n} \sum_{x=1}^n t_x^2$$

the least squares estimate of the trend. Here,

$$Ave(b) = \beta, \quad Var(b) = \frac{\sigma^2}{nD} (1 + F).$$

For the difference,

$$\Delta = b_2 - b_1, \quad Ave(\Delta) = \beta_2 - \beta_1, \\ Var(\Delta) = \frac{\sigma_1^2 + \sigma_2^2}{nD} \left(1 + \frac{\sigma_1^2 F_1 + \sigma_2^2 F_2}{\sigma_1^2 + \sigma_2^2} \right)$$

with

$$F = 4 \sum_{\alpha=1}^{n-1} \left(\frac{(n-\alpha)(n-\alpha+1)(n-\alpha+2)}{n(n+1)(2n+1)} \right) r(\alpha)$$

depending on serial correlation.

Although these formulae formally indicate how to adjust the sample statistics to account for serial correlation, this approach cannot be followed directly in practice since the calculation of C and F from the data is sensitive to sampling variations. To overcome this for the case of large sample sizes (> 30) such as those used everywhere here, ZvS propose to approximate the terms based on the lag 1 autocorrelation coefficient with $r(\alpha) = r_1^\alpha$ as for an AR(1) process. For large n and modest r , the first order approximation is $C \approx F \approx 2r_1$ and this is the level of approximation used here. The t -test of the difference of means is modified as $t = d/S_d \sqrt{(1 + 2r_1)}$ and similarly for the difference of regression coefficients Δ where r_1 is the average lag 1 autocorrelation. The usual sample variance is largely unaffected by serial correlation since the factor $2r/(n-1)$ is small. The approximate statistical tests otherwise use standard formulae as in Crow et al. (1960) or von Storch and Zwiers (1999).

As an illustration, Fig. A1 displays r_1 for zonally average values of temperature, precipitation and mean sea-level pressure for DJF from the control simulation of the model. Values of r_1 are small and scattered around zero for precipitation and mean sea-level pressure. Temperature has a modestly large value of r_1 . The structure of r_1 is reasonably similar between observed and modelled values of temperature and between masked and unmasked model values although there are some differences. The statistical tests performed here that involve temperature are modified to take this serial correlation into account although, in practice, it has only a small effect on the results.

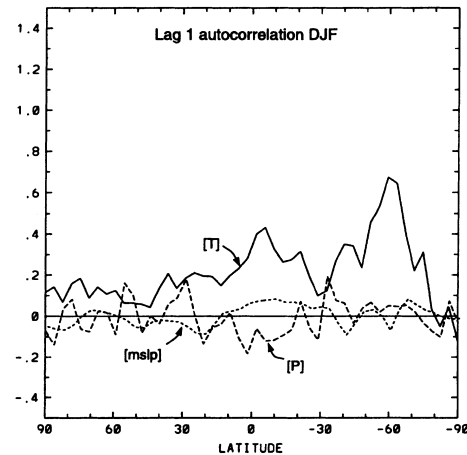


Fig. A.1 Serial correlation in zonally averaged temperature, precipitation, and mean sea-level pressure for the DJF season calculated from the control integration

References

- Boer GJ (1993) Climate change and the regulation of the surface moisture and energy budgets. *Clim Dyn* 8: 225–239
- Boer GJ (1995) Some dynamical consequences of greenhouse gas warming. *Atmos-Ocean* 33: 731–751
- Boer GJ (1999) A study of atmospheric-ocean predictability on long time scales. *Clim Dyn* (in press)
- Boer GJ, McFarlane NA, Lazare M (1992) Greenhouse gas-induced climate change simulated with the CCC second-generation general circulation model. *J Clim* 5: 1045–1077
- Boer GJ, Flato G, Ramsden D (2000) A transient climate change simulation with greenhouse gas and aerosol forcing: projected climate for the 21st century. *Clim Dyn* 16: 427–450
- Crow EL, Davis FA, Maxfield MW (1960) *Statistics manual*. Dover Publications, New York, 288 p
- Cubasch U (ed), Caneill J-Y, Filiberti MA, Hegerl G, Johns TC, Keen A, Parey S, Thual O, Ulbrich U, Voss R, Waszkewitz J, Wild M, van Ypersele JP (1997) *Anthropogenic climate change*, European Commission Report EUR 17466 EN, 73 p
- Flato GM, Boer GJ, Lee W, McFarlane N, Ramsden D, Reader M, Weaver A (2000) The Canadian Centre for Climate Modelling and Analysis global coupled model and its climate. *Clim Dyn* 16: 451–467
- Haywood JM, Stouffer RJ, Wetherald RT, Manabe S, Ramaswamy V (1997) Transient response of a coupled model to estimated changes in greenhouse gas and sulfate concentrations. *Geophys Res Lett*, 24: 1335–1338
- Hulme M (1992) Global land precipitation climatology for the evaluation of general circulation models. *Clim Dyn* 7: 57–72
- Hulme M (1994) Validation of large-scale precipitation fields in general circulation models. In: Desbois M, Desalmand F (eds) *Global precipitation and climate change*. NATO ASI Series, Springer, Berlin Heidelberg New York
- IPCC (1990) *Climate change, the IPCC scientific assessment*. Houghton JT, Jenkins GJ, Ephraums JJ (eds) Cambridge University Press, Cambridge, 364 p
- IPCC (1992) *Climate change 1992, the supplementary report to the IPCC scientific assessment*. Houghton JT, Callander BA, Varney SK (eds) Cambridge University Press, Cambridge, 200 p
- IPCC (1995) *Climate change 1995, the science of climate change*. Houghton JT, Meira Filho LG, Callander BA, Harris N, Kattenberg A, Maskell K (eds) Cambridge University Press, Cambridge, 572 p
- Jones PD (1994) Hemispheric surface air temperature variations: a reanalysis and an update to 1993. *J Clim* 7: 1794–1802
- Kittel TGF, Giorgi F, Meehl GA (1998) Intercomparison of regional biases and doubled CO₂-sensitivity of coupled atmosphere-ocean general circulation model experiments. *Clim Dyn* 14: 1–16
- Lambert SJ (1995) The effect of enhanced greenhouse warming on winter cyclone frequencies and strengths. *J Clim* 8: 1447–1452
- Langner J, Rodhe H (1991) A global three-dimensional model of the tropospheric sulphur cycle. *J Atmos Chem* 13: 225–263
- McFarlane NA, Boer GJ, Blanchet J-P, Lazare M (1992) the Canadian Climate Centre second generation general circulation model and its equilibrium climate. *J Clim* 5: 1013–1044
- Meehl GA, Boer GJ, Covey C, Latif M, Stouffer RJ (1997) Intercomparison makes for a better climate model. *EOS* 78: 445–451
- Mitchell JFB, Johns TC (1977) On modification of global warming by sulphate aerosols. *J Clim* 10: 245–267
- Mitchell JFB, Johns TC, Gregory JM, Tett SFB (1995) Climate response to increasing levels of greenhouse gases and sulphate aerosols. *Nature* 376: 501–504
- Nicholls N et al. (1996) *Observed climate variability and change*. In: Houghton et al. (eds) *Climate change*. Cambridge University Press, Cambridge
- Pacanowski RC, Dixon K, Rosati A (1993) *The GFDL modular ocean model users guide*. GFDL Ocean Group Tech Rep 2. Geophysical Fluid Dynamics Laboratory, Princeton, USA, 46 p
- Parker DE, Folland CK, Jackson M (1995) Marine surface temperature: observed variations and data requirements. *Clim Change* 31: 559–600
- Reader MC, Boer GJ (1998) The modification of greenhouse gas warming by the direct effect of sulphate aerosols. *Clim Dyn* 14: 593–608
- von Storch H, Zwiers F (1999) *Statistical analysis in climate research*. Cambridge University Press, Cambridge
- Zwiers F, Kahren S (1998) Changes in the extremes of the climate simulated by the CCC GCM2 under CO₂ doubling. *J Clim* 9: 2200–2222
- Zwiers F, von Storch H (1995) Taking serial correlation into account in tests of the mean. *J Clim* 8: 336–351